

Session 4 Quiz - Types of Machine Learning

Track A | Grades 5-8

Q1 [Type: MCQ | Difficulty: Easy | QID:56711]

When a machine learns from examples that already have the correct answers attached, it is using _____ learning.

- A. Supervised
- B. Unsupervised
- C. Manual
- D. Reinforcement

Correct Answer: A. Supervised

Explanation: Supervised learning relies on labeled data, where each example has a correct answer attached. This acts like an answer key for the machine to learn from. Unsupervised learning has no labels, reinforcement learning uses trial and error, and manual is not a standard ML type.

Q2 [Type: MCQ | Difficulty: Easy | QID:56712]

In the context of Machine Learning, 'labeled data' is most similar to using _____ to study.

- A. A mystery novel
- B. Flashcards
- C. A video game
- D. A blank notebook

Correct Answer: B. Flashcards

Explanation: Flashcards provide both a question and a correct answer on the back, mimicking how labeled data works. A mystery novel hides answers, a video game involves trial and error (reinforcement), and a blank notebook has no answers.

Q3 [Type: MCQ | Difficulty: Easy | QID:56713]

Unsupervised learning is often compared to the task of _____ because the machine must find its own way to group items.

- A. Following a baking recipe
- B. Taking a multiple-choice test
- C. Training a puppy with treats
- D. Organizing a messy closet

Correct Answer: D. Organizing a messy closet

Explanation: In this analogy, the machine groups similar items together based on their features without being told which category they belong to. Just as you would group shirts with shirts and pants with pants by looking at them, unsupervised learning finds hidden structures on its own.

Q4 [Type: MCQ | Difficulty: Easy | QID:56714]

When an AI learns by trial and error to get the highest possible score, it is using _____ learning.

- A. Reinforcement
- B. Clustering
- C. Unsupervised
- D. Supervised

Correct Answer: A. Reinforcement

Explanation: Reinforcement learning involves the machine performing actions and receiving rewards or penalties to improve its strategy, similar to leveling up in a video game. Clustering is an unsupervised technique, not a learning type.

Q5 [Type: MCQ | Difficulty: Easy-Medium | QID:56715]

The specific job of Supervised Learning that involves predicting a number, such as tomorrow's temperature, is called _____.

- A. Classification
- B. Clustering
- C. Logic
- D. Regression

Correct Answer: D. Regression

Explanation: Regression is the branch of supervised learning specifically used for predicting continuous numerical values like temperature, price, or time. Classification predicts categories (spam or not spam). Clustering is unsupervised, and logic is not an ML task.

Q6 [Type: True/False | Difficulty: Easy | QID:56716]

True or False: In Unsupervised Learning, the machine is given a 'teacher' who identifies all the patterns for it.

- A. True
- B. False

Correct Answer: B. False

Explanation: The word "unsupervised" means there is no teacher. The machine must find hidden structures and groups on its own because there are no labels or answers provided. Supervised learning has a teacher (the labels), not unsupervised.

Q7 [Type: True/False | Difficulty: Easy | QID:56717]

True or False: Regression and Classification are both tasks performed using Supervised Learning.

- A. True
- B. False

Correct Answer: A. True

Explanation: Both regression (predicting numbers) and classification (predicting categories) rely on labeled training data to learn. They are the two main types of supervised learning tasks.

Q8 [Type: True/False | Difficulty: Easy | QID:56718]

True or False: Reinforcement Learning requires a large dataset of labeled photos to begin training.

- A. False
- B. True

Correct Answer: A. False

Explanation: Reinforcement learning often starts with no data and learns by interacting with an environment to receive feedback (rewards or penalties). It does not require labeled photos like supervised learning does.

Q9 [Type: True/False | Difficulty: Easy-Medium | QID:56719]

True or False: A robot learning to walk by falling down and trying again is an example of Unsupervised Learning.

- A. False
- B. True

Correct Answer: A. False

Explanation: This is actually an example of reinforcement learning, where falling is a penalty and walking is a reward. Unsupervised learning is about finding patterns in unlabeled data, not trial and error.

Q10 [Type: True/False | Difficulty: Easy-Medium | QID:56721]

True or False: Real-world AI applications like Netflix often combine multiple types of machine learning at once.

- A. True
- B. False

Correct Answer: A. True

Explanation: Complex systems use different ML types to handle different tasks. Netflix might use unsupervised learning to group users with similar tastes and reinforcement learning to test which recommendations get the most clicks.

Q11 [Type: MCQ | Difficulty: Easy-Medium | QID:56722]

Which of these is a real-world example of Supervised Learning used for classification?

- A. An AI playing millions of games of chess against itself
- B. An email system identifying spam vs. not spam
- C. Amazon grouping customers who bought similar items
- D. A robot vacuum learning the layout of a living room

Correct Answer: B. An email system identifying spam vs. not spam

Explanation: This system sorts emails into categories (spam or not spam) based on previous examples that were explicitly labeled by humans. That is classification, a supervised learning task. Chess playing is reinforcement learning, grouping customers is unsupervised, and robot vacuum mapping is often reinforcement or unsupervised.

Q12 [Type: MCQ | Difficulty: Easy-Medium | QID:56723]

If you wanted to design an AI that groups your messy photo library into 'Photos of People' and 'Photos of Food' without you labeling them first, which style would you use?

- A. Reinforcement Learning
- B. Regression Analysis
- C. Supervised Learning
- D. Unsupervised Learning

Correct Answer: D. Unsupervised Learning

Explanation: Unsupervised learning is perfect for finding similarities and grouping items without needing any pre set labels. The AI would look at visual features and cluster similar photos together. Supervised learning would require you to label thousands of photos first.

Q13 [Type: MCQ | Difficulty: Easy-Medium | QID:56724]

What is the primary goal of Reinforcement Learning?

- A. To memorize an exact recipe provided by a human
- B. To predict a category using labeled data
- C. To get the most rewards through trial and error
- D. To find hidden structures in unlabeled data

Correct Answer: C. To get the most rewards through trial and error

Explanation: Reinforcement learning aims to optimize a strategy by repeating actions that lead to positive outcomes (rewards) and avoiding actions that lead to penalties. It learns from experience, not from labeled data or hidden structures.

Q14 [Type: MCQ | Difficulty: Medium | QID:56725]

Google News uses Unsupervised Learning to group articles. How does it know which articles go together?

- A. It looks for articles that use similar words and patterns
- B. A human editor tells it the topic of every single article
- C. The articles are already numbered by the publishers
- D. It gets a reward every time it puts two random articles together

Correct Answer: A. It looks for articles that use similar words and patterns

Explanation: By noticing patterns in the text (shared keywords, phrases, or named entities), the AI can group similar stories together without needing a human to label them. That is unsupervised learning. Rewards are reinforcement, human labels are supervised.

Q15 [Type: MCQ | Difficulty: Medium | QID:56727]

Why is AlphaGo considered an example of Reinforcement Learning?

- A. It simply groups different Go boards into similar categories
- B. It was trained using millions of labeled Go moves from experts
- C. It uses a set of hard-coded rules written by human players
- D. It learned to play by competing against itself and receiving 'win' rewards

Correct Answer: D. It learned to play by competing against itself and receiving 'win' rewards

Explanation: AlphaGo improved by playing millions of games against itself, treating wins as rewards and losses as penalties. This trial and error with feedback is the core of reinforcement learning. Labeled moves would be supervised, and hard coded rules would be traditional programming.

Q16 [Type: MCQ | Difficulty: Easy-Medium | QID:56728]

Which of these is NOT a characteristic of Supervised Learning?

- A. Uses labeled data
- B. Learns entirely by exploring without any guidance
- C. Requires an answer key during training
- D. Predicts categories or values

Correct Answer: B. Learns entirely by exploring without any guidance

Explanation: Learning entirely by exploring without guidance describes unsupervised learning (or reinforcement learning without rewards). Supervised learning always has guidance in the form of labeled data and an answer key. The other three options are correct characteristics of supervised learning.

Q17 [Type: MCQ | Difficulty: Easy-Medium | QID:56729]

In the 'Baking Cookies' analogy, which method represents Reinforcement Learning?

- A. Buying pre-made cookies from a store
- B. Testing different ingredients and remembering which batch tasted best
- C. Observing cookies and noticing that flat ones are crispy
- D. Following a recipe exactly

Correct Answer: B. Testing different ingredients and remembering which batch tasted best

Explanation: This is trial and error with a "taste" reward, which defines reinforcement learning. Following a recipe exactly is supervised (labeled instructions). Observing patterns is unsupervised. Buying pre-made is not learning.

Q18 [Type: MCQ | Difficulty: Easy-Medium | QID:56730]

Which AI task involves predicting how long a drive will take using Google Maps?

- A. Classification
- B. Reinforcement Learning
- C. Unsupervised Learning
- D. Regression

Correct Answer: D. Regression

Explanation: Predicting a specific number (minutes or hours) falls under regression, which is a supervised learning task. Classification would predict a category like "fast" or "slow". Reinforcement and unsupervised are not used for numerical prediction in this way.

Q19 [Type: MCQ | Difficulty: Medium | QID:56731]

How does Spotify's 'Discover Weekly' use Unsupervised Learning?

- A. By identifying groups of songs that 'go together' based on how they sound
- B. By rewarding the AI when you don't skip a song
- C. By following strict rules set by music experts
- D. By checking a list of genres you specifically typed in

Correct Answer: A. By identifying groups of songs that 'go together' based on how they sound

Explanation: Finding inherent groups (clusters) of songs based on audio features, without being told their genre, is a classic unsupervised task. Rewards would be reinforcement, rules are traditional programming, and typed genres are labels (supervised).

Q20 [Type: MCQ | Difficulty: Easy-Medium | QID:56732]

Which type of learning is best for a robot learning to navigate a room without hitting obstacles by getting a 'penalty' when it bumps into something?

- A. Unsupervised Learning
- B. Data Classification
- C. Reinforcement Learning
- D. Supervised Learning

Correct Answer: C. Reinforcement Learning

Explanation: The penalty for bumping into things allows the robot to learn better strategies through experience. This trial and error with feedback (penalties and rewards) is the definition of reinforcement learning. Unsupervised has no feedback, supervised requires labeled examples of correct paths.